SQQZK4993 ACADEMIC PROJECT(H)

Universiti Utara Malaysia

A PREFERENCE-BASED OPTIMIZATION MODEL FOR UNIVERSITY BADMINTON COURT SCHEDULING USING A GENETIC ALGORITHM

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1.0 INTRODUCTION

Badminton is one of the most popular indoor sports at universities. However, high demand in university indoor courts create challenges in allocating courts fairly and efficiently based on user preferences. As highlighted by Chen et al. (2025), sports facilities often experience changing demand, making simple or fixed scheduling methods become less effective.

This study develops a preference-based optimization model using a Genetic Algorithm (GA) to ensure fair, efficient, and user-centric scheduling.

2.0 PROBLEM STATEMENT

- Existing studys mainly focus on reservation management rather than optimization.
- Few mathematical optimization models exist specifically for badminton court scheduling.
- Most existing studys rarely consider users' preferences.
- The court scheduling problem is a NP-hard problem because it consists of multiple constraints (time, courts, users).
- Manual allocation may lead to:
  - duplicate bookings
  - scheduling conflicts
  - unfair allocation
  - low user satisfaction

3.0 RESEARCH OBJECTIVES

RO1: To maximize user preference satisfaction while ensuring efficient and fair badminton court scheduling by developing and evaluating a preference-based optimization model.
RO2: To identify the objective function and constraints in badminton court scheduling that satisfy user preferences.
RO3: To solve the preference-based court scheduling model using Genetic Algorithm.
RO4: To evaluate the performance of the proposed model based on user preference satisfaction, penalty(conflicts), fitness value and court utilization.

4.0 SIGNIFICANCE OF STUDY

- Maximizes user satisfaction by assigning their preferred playing days and sessions (Morning, Afternoon, Evening).
- Demonstrates a real-world application of Genetic Algorithm in sport scheduling problem.
- Improves operational efficiency and reduces booking conflicts for the university sports centre.

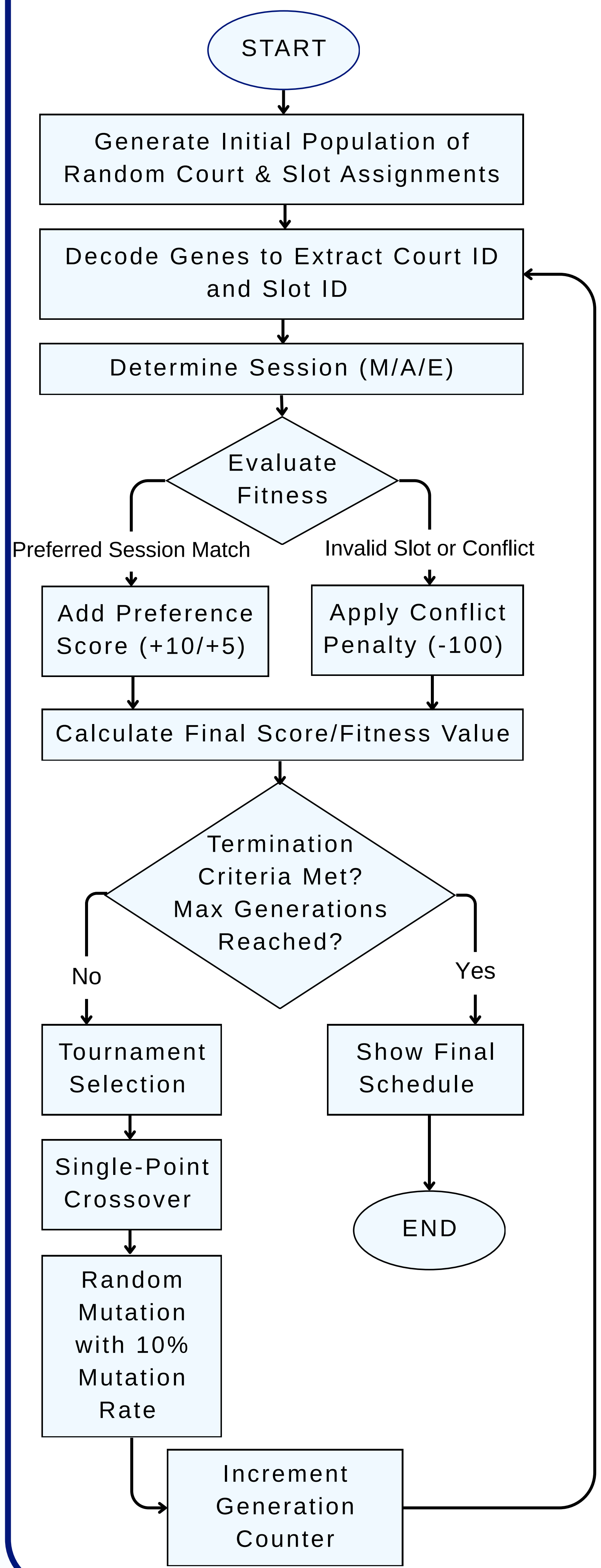
5.0 LITERATURE REVIEW

Theme	Key Findings
Sport Facility Scheduling	Intelligent scheduling improves resource allocation and reduces scheduling inefficiencies in sports facilities (Wang, 2024).
User Preference-Based Scheduling	Incorporating user preferences enhances scheduling quality and user satisfaction, but has rarely been applied to sports facilities (Vemuri, 2024).
Genetic Algorithm (GA)	Genetic Algorithms effectively solve complex NP-hard scheduling problems by generating high-quality solutions within reasonable computational time (Goyal et al., 2026).

Research Gap:
No existing study combines Genetic Algorithm with user preferences especially for university badminton court scheduling.

6.0 METHODOLOGY

6.1 GENETIC ALGORITHM PROCESS



6.2 FITNESS FUNCTION

FITNESS VALUE = TOTAL PREFERENCE SCORE - TOTAL PENALTY

Preference Score
- 1st Preference (Session 1) = +10
- 2nd Preference (Session 2) = +5
- Other Session = 0

Penalty :
Duplicate booking (same day, court and slot) **= -100**

6.3 SCHEDULING DATA

- 6 courts
- 14 slots per day
- 3 sessions per day
  - Morning (Slots 1-4)
  - Afternoon (Slots 5-9)
  - Evening (Slots 10-14)
- 6 operating days per week : (Monday to Sunday, excluding Saturday).
- Total weekly capacity: 504 slots (6 courts x 6 days x 14 slots)
- 5 datasets: 30, 50, 100, 300 and 500 users

6.4 GA PARAMETERS

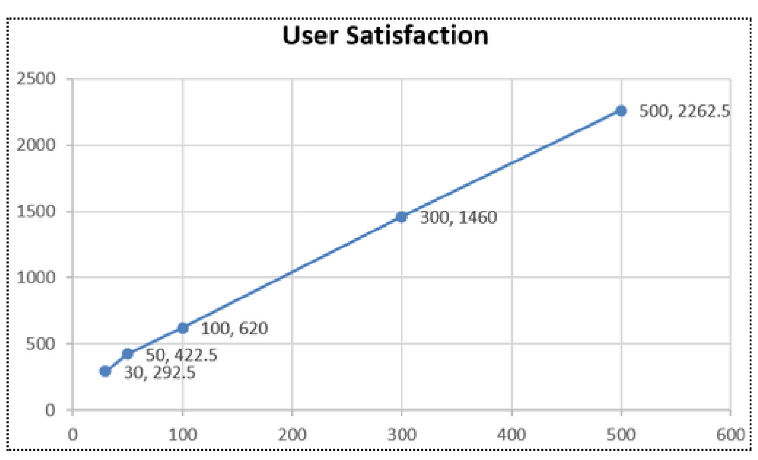
No. of Users	Population Size	No. of Generations
<= 100	40	150
101 - 500	60	200
> 500	80	250

7.0 RESULTS & ANALYSIS

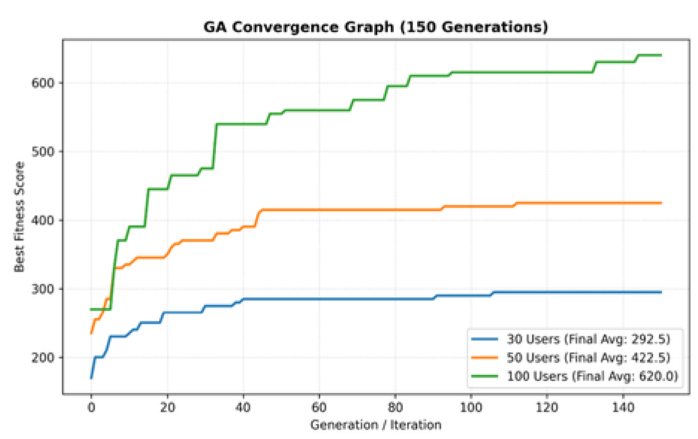
7.1 OVERALL AVERAGE PERFORMANCE SUMMARY

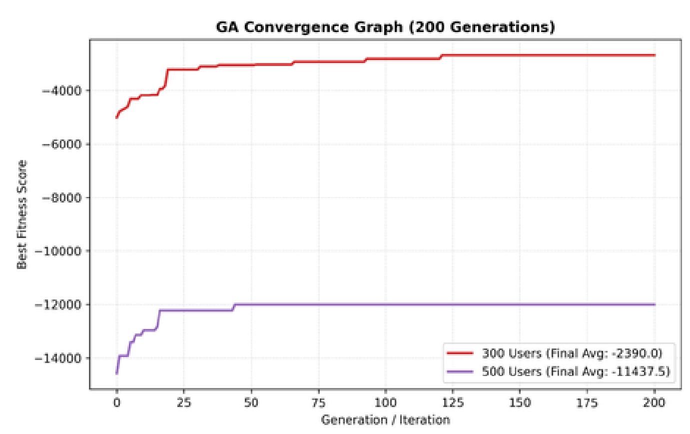
No. of Users	Generations	Total Score	Total Penalty	Final Fitness	Utilization(%)	Conflicts
30	150	292.5	0	292.5	5.95	0
50	150	422.5	0	422.5	9.92	0
100	150	620	0	620	19.84	0
300	200	1460	3850	-2390	59.52	38.5
500	200	2262.5	13700	-11437.5	99.21	137

As the number of users increase beyond the available capacity (504 slots), conflicts occur and penalties increase, leading to negative fitness scores.



7.2 GENETIC ALGORITHM CONVERGENCE





Research Assumptions
- Each user submits one booking request with one preferred day and two preferred sessions.
- The Genetic Algorithm is stochastic. Therefore, results may vary between runs. The reported results represent selected representative runs under the same parameter settings.
- Users assigned to duplicate bookings (if any occur due to capacity limitations) are assumed to accept the generated schedule without requesting manual rescheduling.

7.3 KEY FINDINGS

✓ High user satisfaction was achieved for small and medium datasets.
✓ Conflict-free schedules were generated for 30–100 users.
✓ User satisfaction decreased as booking demand approached court capacity.
✓ The Genetic Algorithm consistently converged to high-quality scheduling solutions.

8.0 CONCLUSION & RECOMMENDATIONS

The proposed Genetic Algorithm (GA) successfully generated efficient and preference-based badminton court schedules by maximizing user satisfaction and minimizing booking conflicts. The model achieved conflict-free scheduling for low and moderate user demand (30–100 users) while higher demand reduced user satisfaction due to limited court capacity. However, the current model relies on static datasets, a single facility setup that is 6 courts and a single-objective optimization model. Future research should explore multi-objective or hybrid optimization algorithms. Furthermore, integrating live, real-time rescheduling and expanding the model to support multi-sport facilities will significantly maximize its real-world utility.